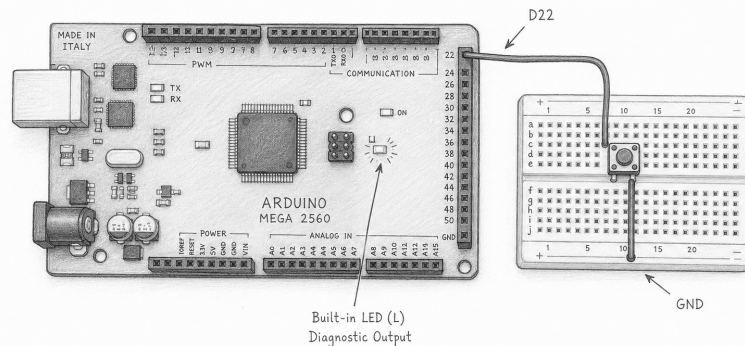


Digital Input

A safe pull-up switch on Arduino Mega 2560

Observe the pin first; interpret human intent second.

Lesson	002	Time	70–95 minutes
Requires	Lesson 001: Digital Output	Board	Arduino Mega 2560
Outcome	Read and diagnose one grounded switch	Evidence	Truth table, LED/test points, host test, three trials



Orientation only. The exact schematic and connection table below govern wiring; verify D22 and GND before applying power.

1 Why input comes this early

An output lets firmware report what it thinks. An input lets a person change what firmware does. Together they form the smallest useful debugging panel. This lesson keeps the two ideas separate: the new component reports the *electrical* pin level without debouncing or inventing button events. Lesson 003 will build those policies on top.

Learning targets

By the end, you can:

- wire a normally-open switch from a Mega input to ground;
- explain why an internal pull-up gives an unpressed pin a known level;
- distinguish a sampled electrical level from an application's meaning;
- diagnose open, stuck-low, and noisy input evidence;
- drive the same input sequence through a deterministic host fake;
- explain how ownership and RAI leave the pin passive at shutdown.

2 Safety and equipment

Energy class E1: one USB-powered Arduino Mega 2560, one dry-contact switch, the board's built-in LED, DMM, and an identified 10 k Ω D13 weak-bias resistor. No external supply or stored-energy load is admitted.

Safety checkpoint. Disconnect USB, barrel power, batteries, programmers, and powered modules before changing wiring. The switch must connect D22 to GND—never to an output. Do not connect 5 V directly to GND. Use only a dry-contact switch in this lesson. Do not use mains-powered, automotive, pyrotechnic, or unknown equipment.

Required: Arduino Mega 2560 Rev3 or identified compatible board, USB data cable, breadboard, normally-open momentary pushbutton, jumper wires, one measured 10 k Ω resistor, the board's built-in LED, and a DMM with documented input resistance and range.

Identity record: board maker/model/revision _____, board marking _____, switch part or kit label _____, continuity-confirmed contact pair _____, USB source identity and 5 V rating _____, and cable identity _____. Confirm that LED_BUILTIN names the on-board diagnostic LED for the selected Mega target. A visually similar four-leg switch is not identified until its open and closed contact pair has been measured with all power removed.

Before power: ☐ switch straddles the breadboard center gap; ☐ one selected switch terminal reaches D22; ☐ its opposite terminal reaches Mega GND; ☐ there is no 5 V jumper; ☐ continuity is open when released and closed when pressed.

3 Predict the electrical behavior

The Mega's internal pull-up is a weak resistor from the input to the board supply. It establishes HIGH when the switch is open. Pressing the switch provides a much lower-resistance path to ground, so the pin reads LOW. No external resistor is required for this exercise.

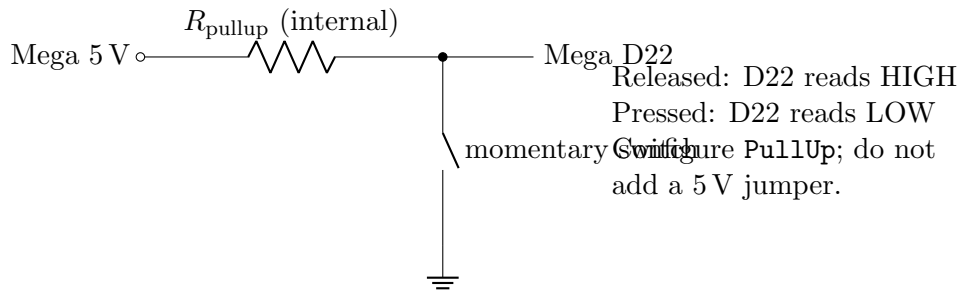
Switch	D22 path	Raw level	Human interpretation
Released	through pull-up to supply	HIGH	switch open
Pressed	through switch to GND	LOW	switch closed

Prediction: If the ground wire falls out, the input will probably remain _____. Why?

Prediction: If D22 is shorted to ground, what will the raw trace show?

Vocabulary boundary. DigitalInput reports Level::High or Level::Low. *Pressed*, *released*, *pressed edge*, and *debounced* belong to a Button component. The application may compare a level with LOW, but DigitalInput does not attach human meaning to it.

4 Build from the exact schematic



From	To	Check with power disconnected
Mega D22	one side of switch	continuous to the chosen leg
opposite switch side	Mega GND	continuous to a GND pin
D22	GND	open released; near zero ohms pressed

Four-leg tactile switches connect the two legs on each side internally. Rotating one by 90° can make the input permanently grounded. Use continuity mode or the switch datasheet; do not trust physical symmetry.

5 Read state deliberately

The canonical `Lesson002DigitalInput.ino` composes the input with Lesson 001's output:

```
constexpr adk::PinId ButtonPin = 22;

adk::Runtime      runtime;
adk::DigitalInput buttonInput (runtime.resources (), ButtonPin, adk::Pull::Up);
adk::MonoLed      led          (runtime.resources (), LED_BUILTIN);

void setup ()
{
    buttonInput.initialize ();
    led.initialize          ();
}

void loop ()
{
    buttonInput.update ();

    const bool pressed = buttonInput.read () == adk::Level::Low;
    led.set (pressed);
}
```

The variable is named `button` because of its circuit role; its type remains `DigitalInput`. `update()` samples hardware and stores the level. `read()` returns that stored level without a surprise hardware access. `sample()` performs an immediate hardware read without changing the stored level. Prefer `update()` plus `read()` for coherent application frames; reserve `sample()` for deliberate diagnostics.

The compact example omits error reporting to expose the central composition. In an application, compare every initialization result explicitly:

```
const adk::Status buttonStatus = button.initialize ();
```

```

const adk::Status ledStatus    = led.initialize    ();

if (!buttonStatus.ok () || !ledStatus.ok ())
{
    button.shutdown ();
    led.shutdown    ();
    return;
}

```

RAII contract: construction stores configuration without touching hardware; successful `initialize()` claims and configures D22; repeated `shutdown()` is harmless; destruction releases the claim and restores a passive input state. Internal code reports failure with `Status`, not exceptions, while cleanup remains correct if surrounding application code throws.

The downloadable canonical sketch begins with a 250 ms acquisition pulse and 750 ms dark separator before input behavior begins. It adds one controlled acceptance action: hold the switch continuously for three seconds to call `shutdown()` on the LED and input. The sketch then remains passive until reset. This is a diagnostic path, not button debouncing and not emergency power removal.

6 Predict, observe, interpret

The canonical sketch does not produce Serial output. Its primary evidence is the built-in LED, driven from the same cached sample used by the application. TP-IN is Mega D22 measured relative to GND; TP-LED is the built-in LED indication. Repeat three voltage/LED trials for each state; the DMM observes TP-IN voltage, while the LED exposes the cached firmware interpretation.

Predict — acquisition: one 250 ms LED pulse, then 750 ms dark, precedes switch behavior.

Observe — acquisition timing: _____

Interpret — acquisition: _____

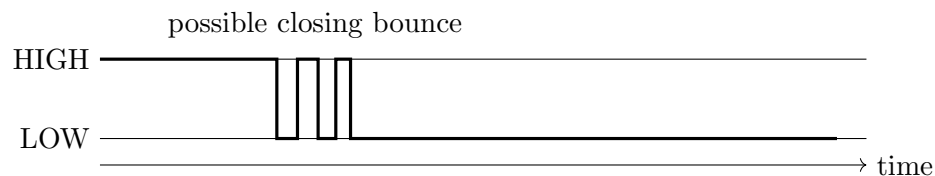
Trial	Switch	Expected level	TP-IN voltage	Diagnostic LED / notes
1	released	HIGH	_____	_____
1	pressed	LOW	_____	_____
2	released	HIGH	_____	_____
2	pressed	LOW	_____	_____
3	released	HIGH	_____	_____
3	pressed	LOW	_____	_____

Diagnostic mapping: the canonical sketch turns the Lesson 001 LED on when `button.read () == adk::Level::Low`. The LED is a view of the stored sample, not a second input read. If TP-IN has the predicted level but the LED is wrong, investigate the output path; if both are wrong, investigate input wiring.

Predict	Observe	Interpret
Released: TP-IN HIGH; LED off	Record TP-IN voltage and LED in three trials	Agreement supports the open-contact and pull-up paths
Pressed: TP-IN LOW; LED on	Record TP-IN voltage and LED in three trials	Agreement supports the closed contact-to-GND path
Reset while held: LOW and LED on after startup	Hold, tap reset, then observe without rewiring	Initialization sampled the real startup level rather than assuming HIGH
Three-second hold: LED extinguishes and sketch stays passive	Keep the switch held until shutdown, release it, and observe no LED response	The controlled path ran; physical pin-mode evidence is recorded separately
USB removed: no powered indication	Disconnect USB and confirm the board is unpowered before touching wiring	Physical power removal, not software cleanup, makes rewiring permissible

7 Contact bounce is evidence, not yet a bug

A mechanical contact may alternate HIGH–LOW several times while closing. At a 100 ms reporting interval you may miss this; at a faster diagnostic interval you may see it. `DigitalInput` must preserve raw evidence. It must not silently delay, filter, or synthesize events.



Lesson 003 will consume raw samples at explicit times, require stability for a configured interval, and expose button events. Keeping that policy in a separate component allows other users—limit switches, jumpers, and digital sensors—to observe unfiltered levels.

8 Deterministic host test

The API and lifecycle below are **host-verified**. The physical circuit remains **experimental** until you complete and record the Mega acceptance checks. Host tests prove component rules without USB, wall-clock delays, or nondeterministic contact timing.

```
fake::reset ();
adk::ResourceRegistry resources;
fake::setDigitalInput (22, HIGH);
adk::DigitalInput input (resources, 22, adk::Pull::Up);

require (input.initialize ().ok ());
require (input.read () == adk::Level::High);

fake::setDigitalInput (22, LOW);
require (input.sample () == adk::Level::Low);
require (input.read () == adk::Level::High);
```

```
input.update ();
require      (input.read () == adk::Level::Low);

input.shutdown ();
input.shutdown ();
```

This mirrors the repository's fake-Arduino tests: `sample()` observes LOW without mutating the cached HIGH; `update()` then stores LOW. Separate tests verify the exact `pinMode`, `digitalRead`, claim-conflict, invalid-pin, repeated-initialization, repeated-shutdown, and destructor traces. A conflict is reported as a status:

```
adk::DigitalInput owner  (resources, 22, adk::Pull::Up);
adk::DigitalInput blocked (resources, 22, adk::Pull::Up);

require (owner.initialize ().ok ());
require (blocked.initialize ().error () == adk::StatusCode::ResourceBusy);
```

Preserve these assertions when extending the tests:

1. construction alone produces no hardware call;
2. initialization claims D22, selects INPUT_PULLUP, and samples;
3. repeated successful initialization produces no hardware call;
4. `sample()` reads hardware without changing `read()`;
5. `update()` reads hardware and changes `read()`;
6. shutdown is idempotent and restores INPUT;
7. destruction releases the claim;
8. a second owner reports `error() == StatusCode::ResourceBusy`;
9. an invalid pin reports `error() == StatusCode::InvalidPin` without touching hardware.

Extend the test: script HIGH, LOW, HIGH, LOW without elapsed time. Record the four results. Why must DigitalInput preserve all four?

9 Fault laboratory

Change one condition at a time, with power removed before each wiring change. Restore and recheck the exact schematic after every trial.

Safe fault	Expected evidence	What the evidence distinguishes
Disconnect switch-to-GND wire	always <code>Level::High</code>	open return path versus a working press
Move input configuration from pull-up to floating input	unstable or environment-dependent level	defined bias versus a floating conductor; restore immediately
Hold switch pressed during reset	<code>Level::Low</code> after initialization	startup state is sampled, not assumed
Temporarily slow reporting to 500 ms	fewer observations, same mapping	sampling/display rate versus electrical meaning

Safety checkpoint. Do not create faults by shorting an output, supply, or external signal. The only wiring fault used here is an open connection. A floating-input demonstration is brief, uses only D22 and the switch, and is restored to pull-up configuration before continuing.

10 Mega 2560 acceptance checklist

Date/time: _____ operator: _____ reviewer/date: _____
 Board maker/model/revision/ID: _____
 Port: _____
 ADK/sketch commit: _____ upload command: _____
 Arduino CLI: _____ AVR core: _____ FQBN: _____
 USB source/cable/current limit: _____ measured 5 V rail: _____
 Switch identity/contact map: _____ 10 k Ω measured value/rating: _____
 DMM model/range/input resistance: _____

- ☐ Clean build succeeds with warnings treated according to project policy.
- ☐ Sketch size is recorded: flash _____ bytes; SRAM _____ bytes.
- ☐ A distinct 250 ms LED acquisition pulse and 750 ms dark separator occur before switch behavior.
- ☐ Released TP-IN voltage and LED-off agree in three trials.
- ☐ Pressed TP-IN voltage and LED-on agree in three trials.
- ☐ Diagnostic LED agrees with the same stored sample.
- ☐ Three press/release trials match the prediction table.
- ☐ Ground-wire-open fault is detected from evidence and restored.
- ☐ Reset while pressed reports the actual pressed state.
- ☐ A continuous three-second hold enters the passive state; release produces no further LED response until reset.
- ☐ Host trace confirms controlled shutdown sets D22 and LED_BUILTIN to INPUT (high impedance).
- ☐ A named DMM is connected TP-IN-to-GND while unpowered; its input resistance and the exact Mega pull-up limits are recorded.
- ☐ Released TP-IN voltage with the pull-up active and after shutdown is recorded against calculated ranges, separately proving the weak meter path controls the released pin after shutdown.
- ☐ The measured 10 k Ω weak pull-up is installed D13-to-5 V while unpowered. The exact board D13 path is included in predicted driven-LOW and released-pin voltage ranges.
- ☐ The sketch holds D13 LOW for 250 ms before shutdown; its later weak-bias voltage separately supports D13 high impedance.
- ☐ Reset after shutdown reacquires both pins and restores the released/pressed LED mapping.
- ☐ USB is disconnected before the circuit is dismantled.

Deviations and remaining uncertainty: _____

Result: ☐ draft ☐ rejected ☐ hardware verified Reviewer initials: _____

Safety checkpoint. High impedance is a software pin condition, not power isolation. Do not rewire after shutdown while USB or any other source remains connected. Stop for heat, odor, resets, damaged insulation, an unidentified switch, or evidence that disagrees with the prediction. Remove USB before investigating.

11 Explain the result

Claim: In one sentence, state whether the pull-up input distinguishes the two switch states.

Evidence: Cite at least three observations, including measured TP-IN voltage, expected digital level, and the diagnostic output.

Reasoning: Use the pull-up and ground paths to explain why LOW means the switch is pressed in this circuit.

Limit: Why do these trials not prove that every brief raw transition is a deliberate human press?

12 Further work and references

Try one extension at a time:

- record a high-rate raw trace and mark possible bounce;
- use a suitable external pull-down and redraw the exact circuit before building;
- test claim conflicts by attempting two DigitalInput owners for D22;
- replace the switch with a documented, 5 V-compatible digital sensor;
- pass raw snapshots to the Button lesson without changing DigitalInput.

Use the HTML lesson for current repository commands, source links, API signatures, and errata. This printable field guide emphasizes wiring, predictions, evidence, and acceptance. Useful primary references:

- Arduino Mega 2560 Rev3 documentation and pinout: Arduino Mega 2560
- Arduino digital pin and pull-up tutorial: Digital Input Pull-Up
- ATmega2560 datasheet, I/O ports and pull-up behavior: Microchip ATmega2560
- ADK repository documentation: <https://spincyc.github.io/adk/>

Next: Lesson 003 composes DigitalInput with explicit time to create a debounced Button. After that lesson, the first interval project combines digital output, digital input, and button events in a deterministic reaction timer.